MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (BICTU)

SUPPLIMENTARY/DIFFERED EXAMINATION

ICT 4703 :NEWORK ADMINISTRATION AND INFORMATION SECURITY

TIME ALLOWED: 3 HOURS **DATE :NOVEMBER , 2017**

INSTRUCTIONS

i. This paper contains **THREE**pages. Please check.
ii. Answer **ALL** questions.
iii. Marks for each question are indicated.

SECTION A : STRUCTURED QUESTIONS **[50 Marks]**

1. Remote access is often used by system administrators in large organisations to remotely control the computers on the organisation's local area network. Explain how malicious remote access administration tools (RATs) can be used. **[10 Marks]**

2. Is cryptography security? **[10 Marks]**

3. You work in a multi-branched company. You have been given a task to assist management in trimming unnecessary ICT-related costs (Travel expenses, phone) as much as possible. As a Network Administrator with sound and updated knowledge, what would you do? **[10 Marks]**

4. Explain how Cryptography achieves the security goals. **[10 Marks]**

5. Suppose Alice wants to send "Sweetheart" to Bob using vigenere polyalphabetic cipher using key=ENOCH. Calculate the cipher text that Bob will receive. **[10 Marks]**

The Vigenère Square

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
26	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

SECTION B : NETWORK DESIGN **[50 Marks]**

TechNix Malawi Ltd. Network requirements

Youhavebeengiventhenetworkaddress10.51.0.0/24to configureandprovidetheIP addressingforthenetworkshowninFigure 2.Thenetworkhasthefollowingaddressing requirements:

- TheLILONGWELAN1willrequire20hostIPaddresses.
- TheLILONGWELAN2willrequire 15 hostIPaddresses.
- TheMZUZULAN1willrequire30hostIPaddresses.
- TheMZUZULAN2willrequire10hostIPaddresses.
- TheHQ-BTLANwillrequire50hostIPaddresses.
- Mozambique LAN will require 49 Addresses.
- Tanzania LAN will require 27 Addresses.
- The links between routers will require an IP Address at each end.

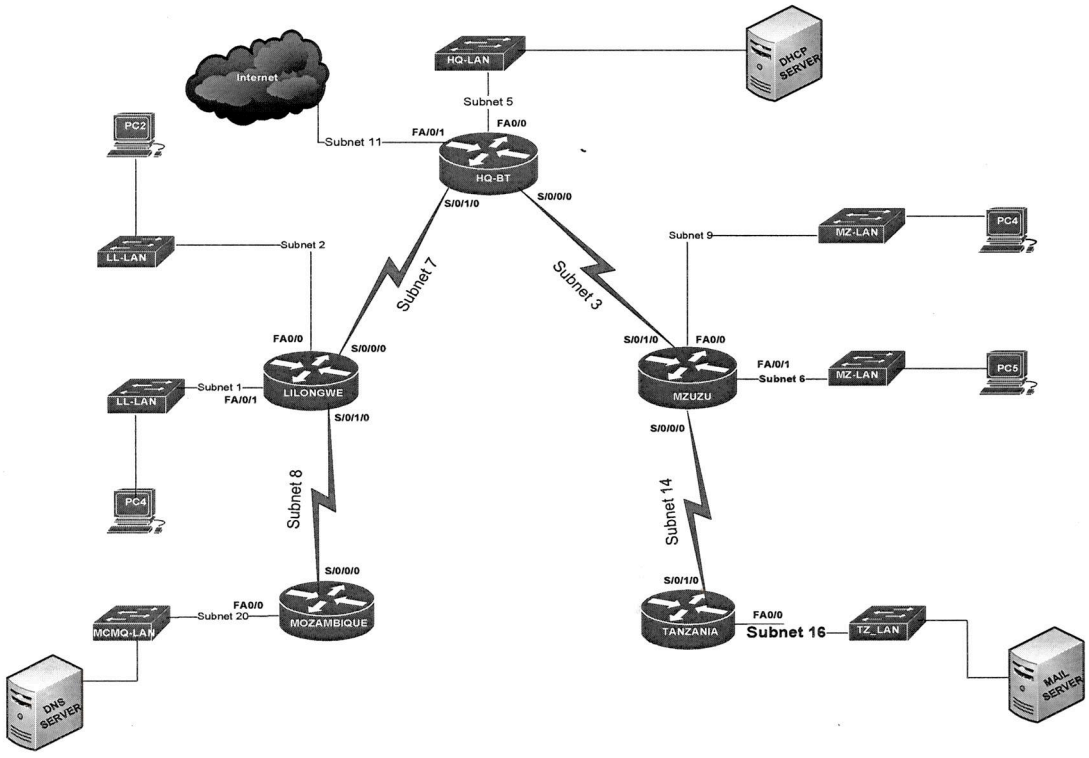


Figure 2. TechNix Malawi Ltd. Network Infrastructure

a) Perform basic configuration on the routers and switches including MOTD. **[10 Marks]**

b) Assigntheappropriateaddressestothedeviceinterfaces using the following instructions: **[15 Marks]**

   i. Assign the first valid address to the serial interfaces s0/0/0 and the second valid address to S0/1/0 on each subnet between routers.

   ii. Assign the last valid address to all PCs.

   iii. Assign the third valid address to all the servers.

c) Implement Static routing on the network. **[10 Marks]**

d) Cofigure the default route on each router **[5 Marks]**

e) What command would enable OSPF? **[5 Marks]**

f) How would you backup config on a remote tftp server with an address 10.168.0.100? **[5 Marks]**

END OF QUESTION PAPER